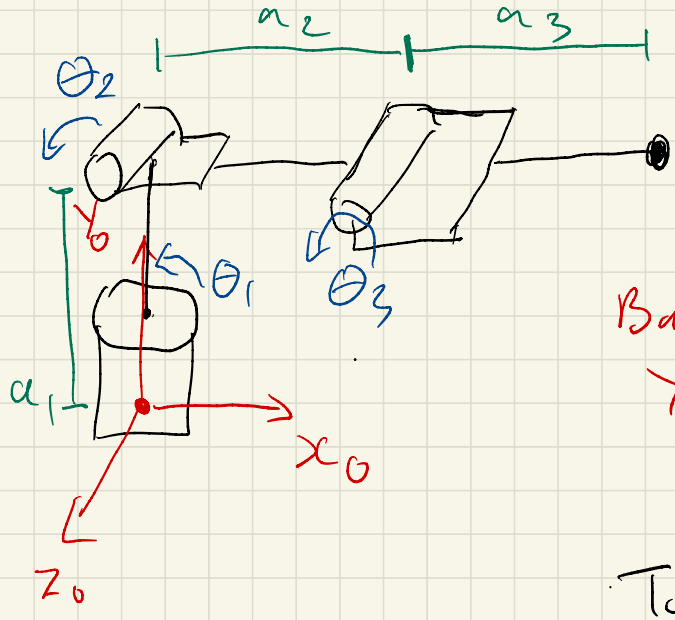


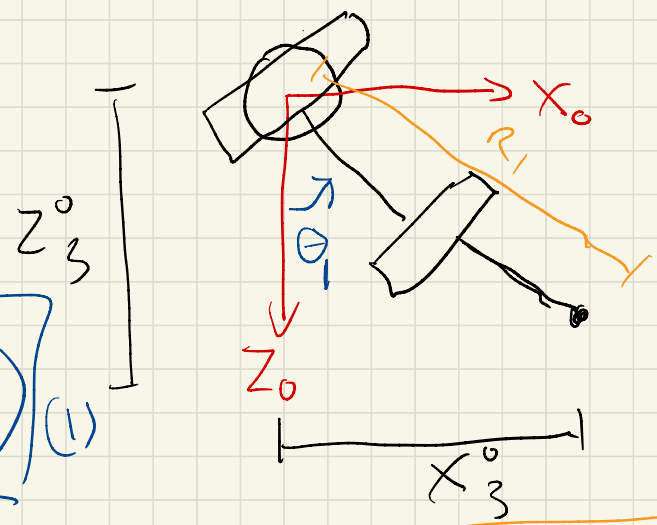
Endeffector Position

(x_3^0, y_3^0, z_3^0)

Base Frame Coordinate

x_0, y_0, z_0

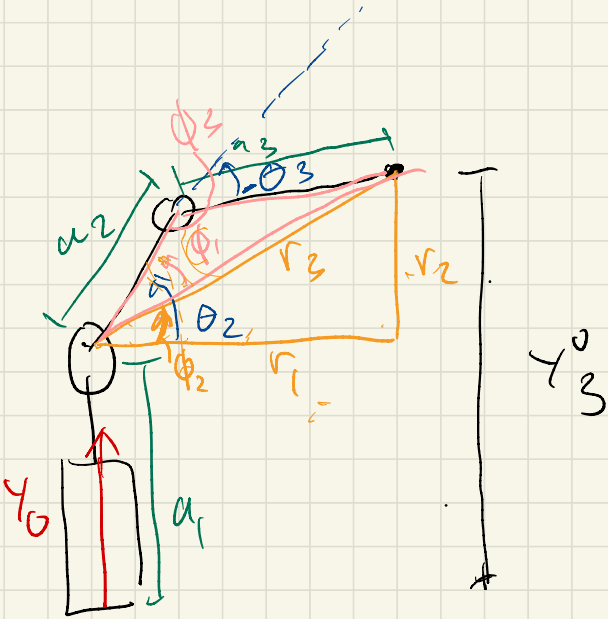
Top View



$$\boxed{\theta_1 = \tan^{-1}\left(\frac{x_3^0}{z_3^0}\right)} \quad (1)$$

$$\boxed{r_1 = \sqrt{(x_3^0)^2 + (z_3^0)^2}} \quad (2)$$

Side View



$$\boxed{\theta_2 = \phi_1 + \phi_2} \quad (7)$$

$$\boxed{r_2 = Y_3^0 - a_1} \quad (3)$$

$$\boxed{\phi_2 = \tan^{-1}\left(\frac{r_2}{r_1}\right)} \quad (4)$$

$$\boxed{r_3 = \sqrt{r_1^2 + r_2^2}} \quad (5)$$

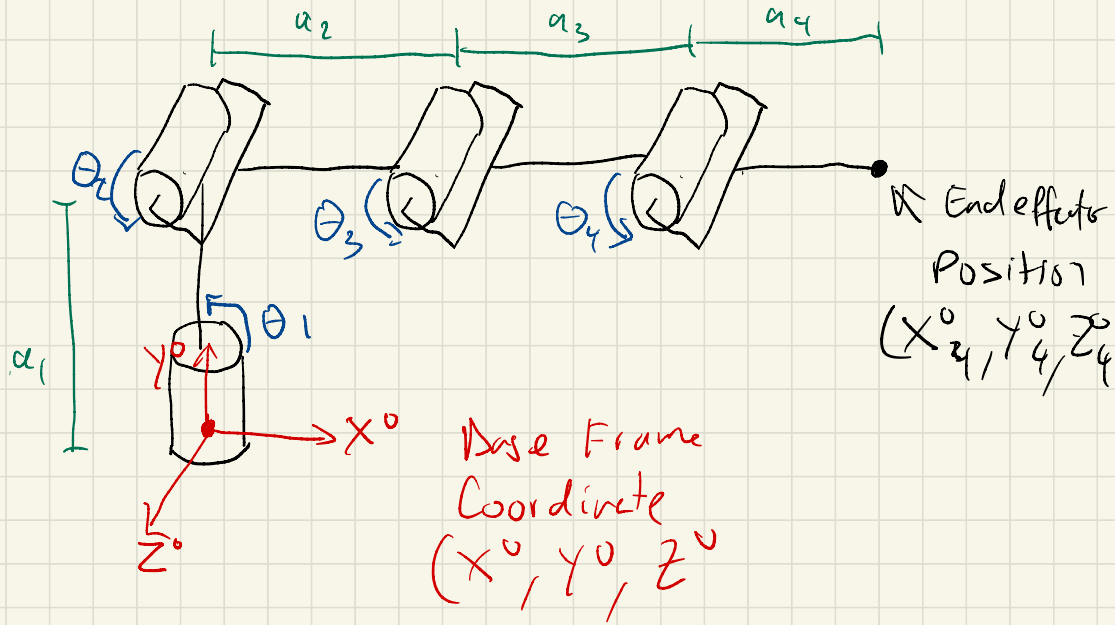
$$a_3^2 = a_2^2 + r_3^2 - 2a_2 a_3 \cos\phi_1$$

$$\boxed{\phi_1 = \cos^{-1}\left(\frac{a_3^2 - a_2^2 - r_3^2}{-2a_2 r_3}\right)} \quad (6)$$

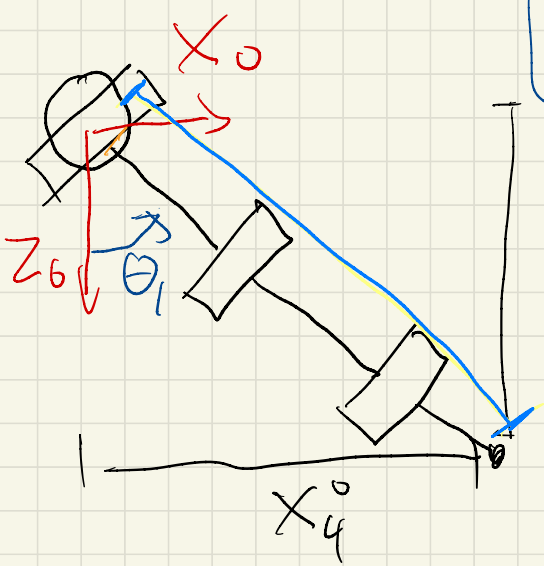
$$\boxed{\theta_3 = \pi - \phi_3} \quad (9)$$

$$r_3^2 = a_2^2 + a_3^2 - 2a_2 a_3 \cos\phi_3$$

$$\boxed{\phi_3 = \cos^{-1}\left(\frac{r_3^2 - a_2^2 - a_3^2}{-2a_2 a_3}\right)} \quad (8)$$



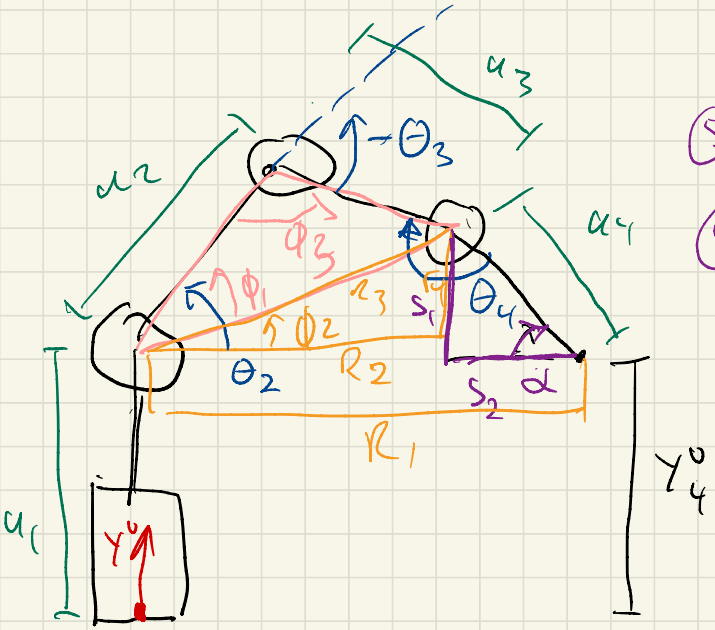
Top View



$$\theta_1 = \tan^{-1}\left(\frac{X^0_4}{Z^0_4}\right) \quad (1)$$

$$M = \sqrt{(X^0_4)^2 + (Z^0_4)^2} \quad (2)$$

α = pitch



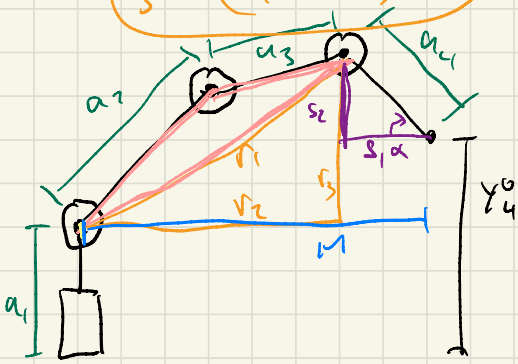
(5) $\boxed{s_2 = a_4 \cdot \sin(\alpha)}$

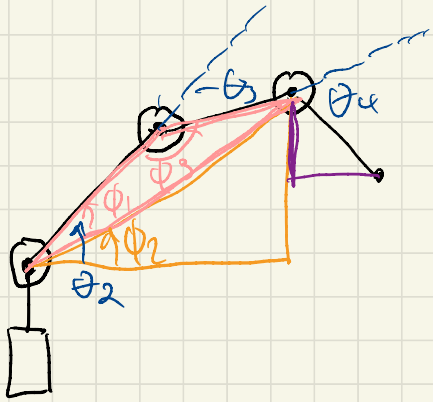
(4) $\boxed{s_1 = a_4 \cdot \cos(\alpha)}$

(6) $r_2 = M - s_1$

(7) $r_3 = (Y_4^0 + s_2) - a_1$

8 $r_1 = \sqrt{r_2^2 + r_3^2}$





ϕ_1